

Runwise Developer Guide

Stack

Prerequisites

- Node.js 22 or later
 - npm
-

Setup

```
git clone https://github.com/alanwaddington/runwise.git
cd runwise
npm install
npm run dev
```

Open <http://localhost:5173>.

Project Structure

```
src/
% % % app.css          # Global styles + Tailwind theme tokens
% % % app.html         # SvelteKit HTML shell
% % % lib/
% % % % affiliates.ts  # Affiliate product definitions per route (Amazon Associates)
% % % % seo.ts         # SEO metadata map (PAGES), sitemap config, OG images
% % % % components/    # Shared UI components
% % % % % AdUnit.svelte      # Consent-gated Google AdSense ad unit
% % % % % AdUnit.test.ts
% % % % % AffiliateLinks.svelte # Per-route affiliate product cards (Amazon)
% % % % % AffiliateLinks.test.ts
% % % % % CookieBanner.svelte # GDPR cookie consent banner (fixed bottom)
% % % % % CookieBanner.test.ts
% % % % % HeroSection.svelte
% % % % % HeroSection.test.ts
% % % % % InputField.svelte
% % % % % InputField.test.ts
% % % % % ResultDisplay.svelte
% % % % % ResultDisplay.test.ts
% % % % % SeoHead.svelte     # Per-page meta tags, OG, JSON-LD, AdSense account
verification
% % % % % SeoHead.test.ts
% % % % % SiteFooter.svelte  # Footer with Privacy Policy link and Manage Cookies
button
% % % % % SiteFooter.test.ts
% % % % % SiteNav.svelte
% % % % % SiteNav.test.ts
% % % % % ToolCard.svelte
% % % % % ToolCard.test.ts
% % % % % ToolLayout.svelte
% % % % % ToolLayout.test.ts
% % % % % stores/           # Svelte stores for cross-component state
% % % % % consent.ts        # GDPR consent read/write (localStorage)
% % % % % consent.test.ts
% % % % % consentBannerVisible.ts # Writable store: true = show banner
% % % % % utils/           # Pure utility modules (no Svelte dependency)
% % % % % pace.ts          # Pace/speed conversion functions
% % % % % pace.test.ts
```

```

%           % % % race-predictor.ts      # Riegel formula, time parsing/formatting, prediction
table
%           % % % race-predictor.test.ts
%           % % % training-paces.ts      # VDOT calculation (Daniels' formula), training zone
pace derivation
%           % % % training-paces.test.ts
%           % % % hr-zones.ts            # Max HR zones, Friel LTHR zones, LTHR sub-zones,
Tanaka age estimate
%           % % % hr-zones.test.ts
%           % % % vo2max.ts              # ACSM normative data, getFitnessCategory,
getAcsmTable, CATEGORY_COLOURS
%           % % % vo2max.test.ts
%           % % % parkrun.ts             # Reference distance list, Riegel prediction, split
generation, PB comparison, WMA age grading
%           % % % parkrun.test.ts
% % % routes/
% % % +layout.svelte    # Root layout — CookieBanner + header + main + SiteFooter
% % % +page.svelte      # Home page — HeroSection + ToolCard grid
% % % +error.svelte     # Error page
% % % pace/
% % % race-predictor/
% % % training-paces/
% % % hr-zones/
% % % vo2max/
% % % parkrun/
% % % privacy/          # Privacy Policy page

```

Design System

Tokens (`src/app.css`)

Dark mode is applied automatically via prefers-color-scheme. Always use design tokens (text-ink, bg-bg, border-ink/10) rather than hardcoded Tailwind colours where tokens exist.

Text Contrast (WCAG AA)

Secondary text (help hints, labels, table headers, footer) uses text-gray-600 (#4b5563), which achieves ~6.4:1 contrast against the light background (#fafaf8) and passes WCAG 2.1 AA (4.5:1 minimum).

An ESLint rule in eslint.config.js errors on text-gray-400 and text-gray-500 in Svelte files. The rule uses a negative lookbehind so dark:text-gray-400 (which passes AA against #19191a) is correctly exempted.

Components

Adding a New Tool

```

Create `src/routes/<tool-name>/+page.svelte`
Import `ToolLayout` and pass `title`, `description`, and `route` props
Add the tool to the `tools` array in `src/routes/+page.svelte`
Add the tool link to `SiteNav.svelte`
Register the route in `src/lib/seo.ts` (`PAGES` map) with a title, description, OG image path, and sitemap
priority — this drives the page's meta tags, `sitemap.xml` entry, `robots.txt` allow rules, and JSON-LD
structured data
Add tests alongside the page

```

Testing

Unit and component tests (Vitest)

```

npm run test          # Run all tests once
npm run test -- --watch # Watch mode

```

Tests live alongside their source files (*.test.ts). Follow the existing pattern:

```
import { describe, it, expect, afterEach } from 'vitest';
import { render, cleanup, screen } from '@testing-library/svelte';
import MyComponent from './MyComponent.svelte';

afterEach(() => { cleanup(); });

describe('MyComponent', () => {
  it('renders correctly', () => {
    render(MyComponent, { props: { ... } });
    expect(screen.getByRole('...')).toBeInTheDocument();
  });
});
```

E2E tests (Playwright)

```
npx playwright install chromium # First time only
npm run test:e2e                # Run E2E tests against the production build
```

E2E tests live in the `e2e/` directory. They run against the production preview server (`npm run build && npm run preview` on port 4173). Configuration is in `playwright.config.ts`.

Code Quality

```
npm run check      # TypeScript / svelte-check
npm run lint       # ESLint
npm run format     # Prettier (auto-fix)
npm run test:e2e   # Playwright E2E tests (requires built app)
```

Workflow

This project follows an issue-driven workflow:

```
/analyse <issue>    !' requirements + acceptance criteria
/design <issue>     !' architecture + work breakdown
/develop <issue>    !' TDD implementation
/verify <PR>        !' runtime verification
/pr-reviewer <PR>  !' acceptance criteria audit
/merge <PR>         !' merge + close issues
```